

Problem 9.4

Express $i^{(6)}$ as a function of $d^{(2)}$.

Solution

The nominal interest rate convertible 6 times per year and the nominal discount rate convertible 2 times per year must produce the same annual accumulation. Thus,

$$\left(1 + \frac{i^{(6)}}{6}\right)^6 = \left(1 - \frac{d^{(2)}}{2}\right)^{-2}.$$

Taking the sixth root of both sides gives

$$1 + \frac{i^{(6)}}{6} = \left(1 - \frac{d^{(2)}}{2}\right)^{-2/6}.$$

Since $-2/6 = -1/3$,

$$1 + \frac{i^{(6)}}{6} = \left(1 - \frac{d^{(2)}}{2}\right)^{-1/3}.$$

Solving for $i^{(6)}$,

$$\begin{aligned}\frac{i^{(6)}}{6} &= \left(1 - \frac{d^{(2)}}{2}\right)^{-1/3} - 1, \\ i^{(6)} &= 6 \left[\left(1 - \frac{d^{(2)}}{2}\right)^{-1/3} - 1 \right].\end{aligned}$$